Descriptive Statistics

Anaya Scott

Question 1

0/1 pt 3 19

Ages	Number of students
15-18	10
19-22	5
23-26	6
27-30	6
31-34	7
35-38	6

Based on the frequency distribution above, find the relative frequency for the class with lower class limit 15

Relative Frequency = ______ %

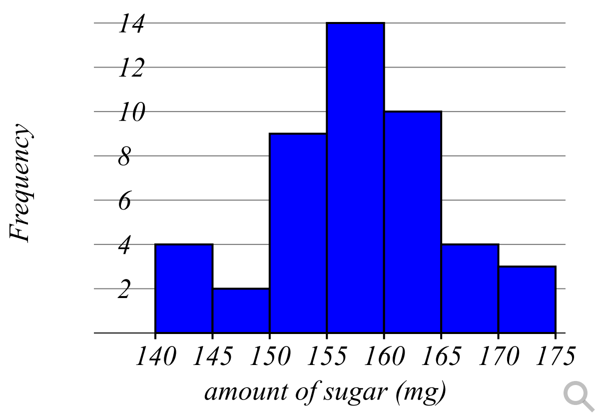
Give your answer as a percent, rounded to one decimal place

Question Help: Video

Question 2

0/1 pt 3 19

Data was collected for a sample of organic snacks. The amount of sugar (in mg) in each snack is summarized in the histogram below.



What is the sample size for this data set?

$n =$ ______

Question Help: Video

Question 3

0/1 pt 3 19

Ages	Number of students
15-18	8
19-22	8
23-26	6
27-30	8
31-34	4
35-38	8

A total of 42 people were sampled. Based on the frequency distribution above, find the degrees in a pie graph for the class with lower class limit 19

Degrees in this section of a pie graph = Give answer to nearest whole number.

Question 4

0/1 pt 3 19

The class ranks of a very small high school with 20 students are shown in the table below.

Fr
Fr
Fr
So
So
So
So
Sr
Jr
Sr
So
Fr
Fr
Fr
Jr
Jr
Sr
Sr
Sr
Fr

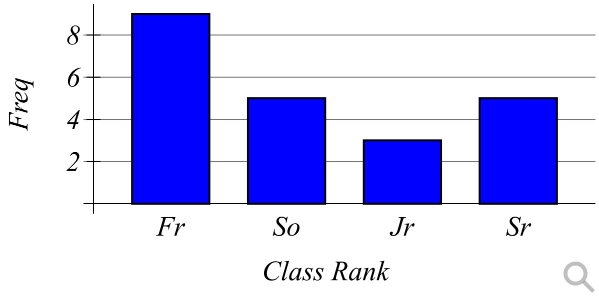
---

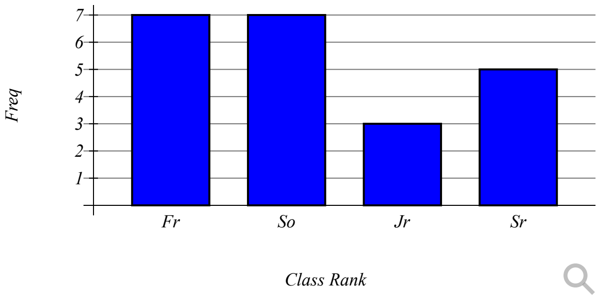
a) Since data were collected for ? Select an answer variable(s), the correct graph to make is a Select an answer .

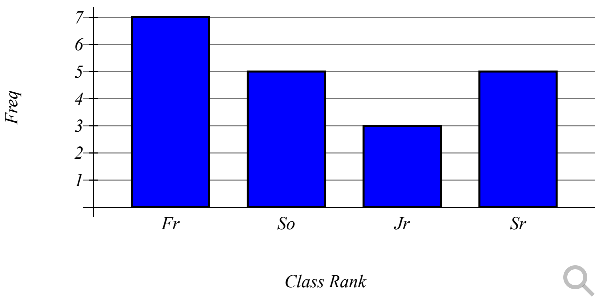
b) Complete the frequency/relative frequency table.

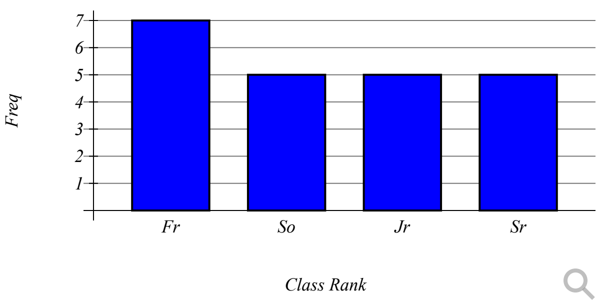
Class Rank	Frequency	Relative Frequency
Fr		
So		
Jr		
Sr		

c) Which of the following is the correct bar chart for the given data?









d) Which of the following is the correct pie chart for the given data?

Question Help: Video 1 Video 2 Video 3 Video 4 Video 5 Video 6

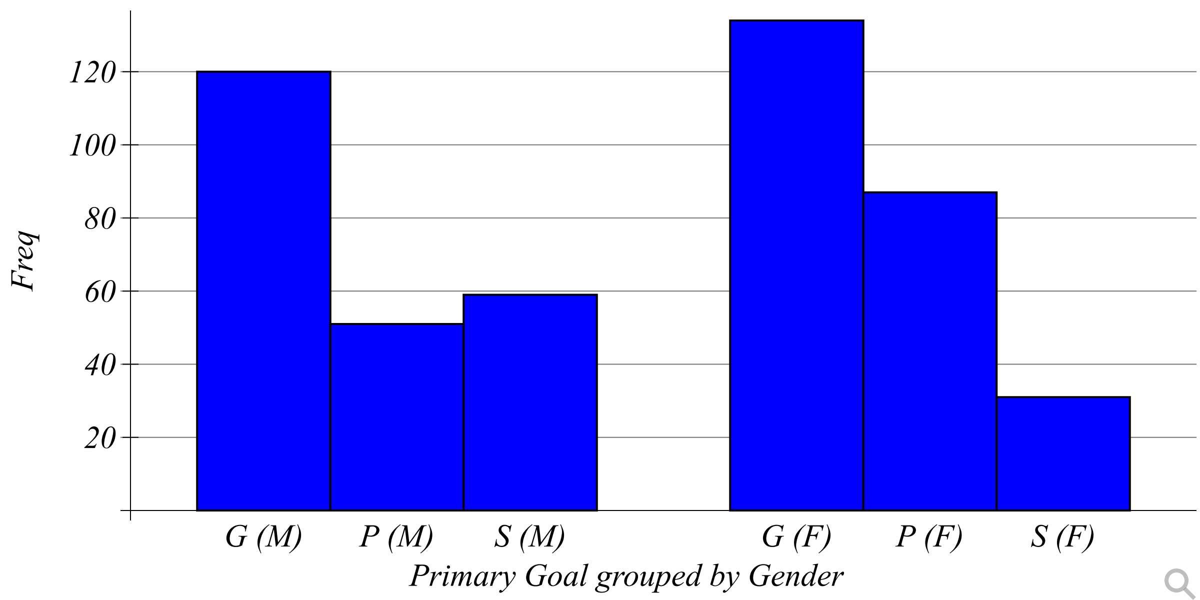
Question 5 0/1 pt 3 19

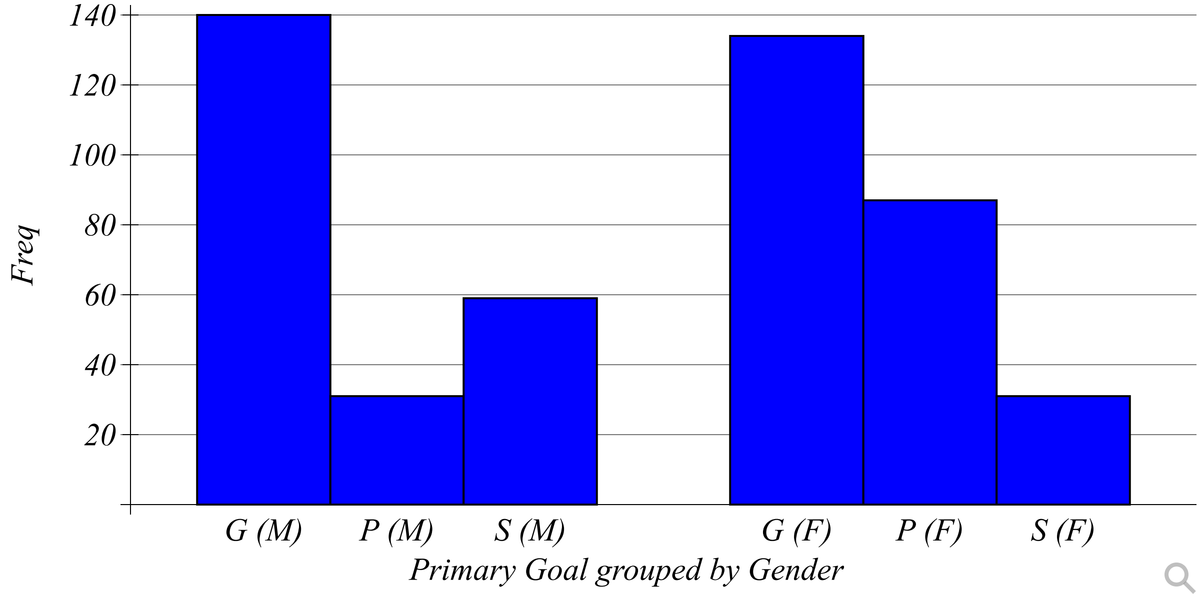
In a study of fourth, fifth and sixth graders, the following data were collected on their gender and their primary goal.

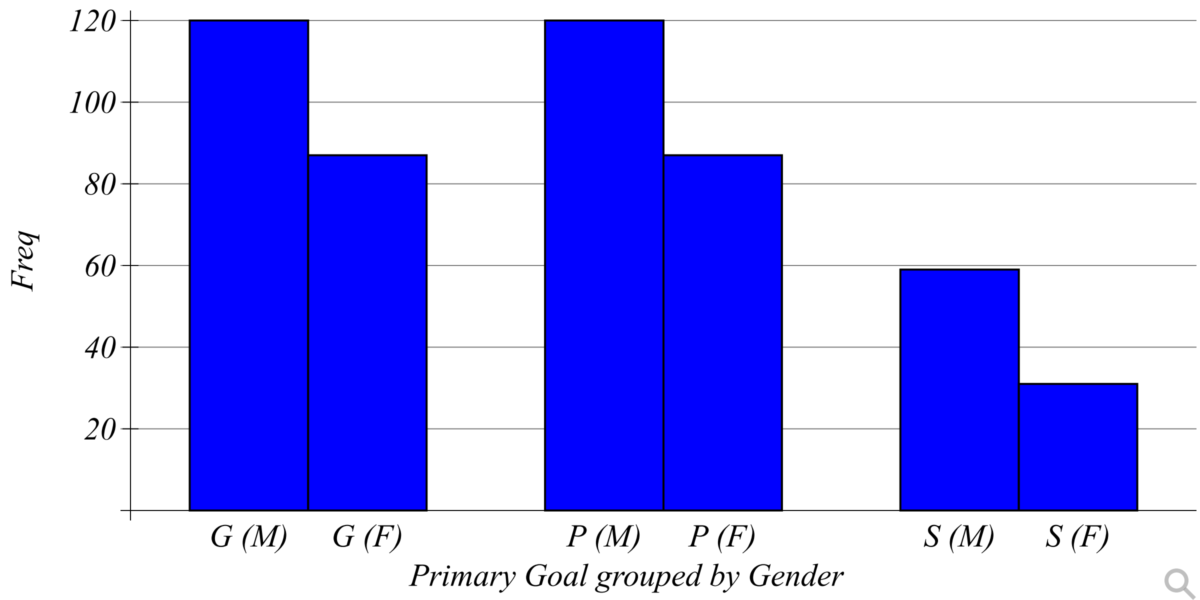
		Primary Goal		
Gender	Good Grades (G)	Popularity (P)	Good at Sports (S)	Row Totals
Male (M)	120	51	59	230
Female (F)	134	87	31	252
Column Totals	254	138	90	482

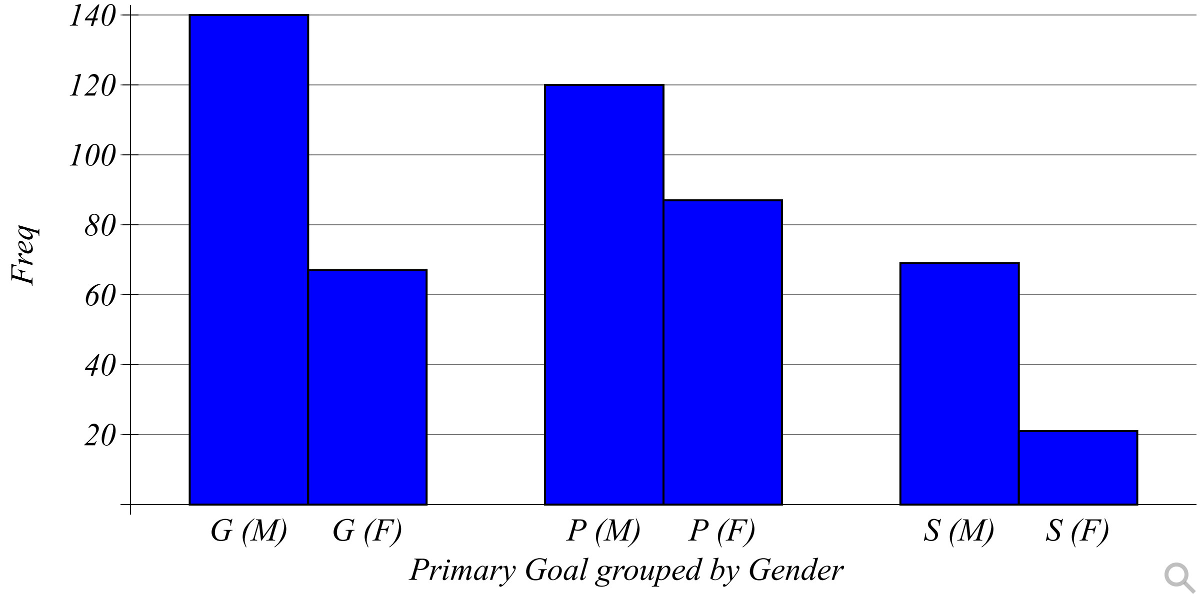
a) Since data were collected for ? Select an answer variable(s), the correct graph to make is a Select an answer

b) Which of the following is the correct graph for the above data with Primary Goal grouped by Gender.




140
120
100
80
60
40
20
Freq
G (M)
P (M)
S (M)
G (F)
P (F)
S (F)
Primary Goal grouped by Gender


120
100
80
60
40
20
Freq
G (M)
G (F)
P (M)
P (F)
S (M)
S (F)
Primary Goal grouped by Gender



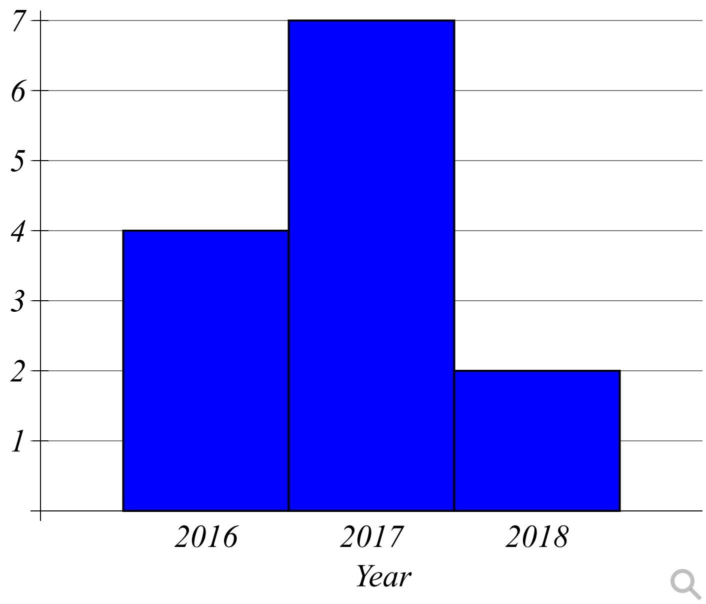
Question Help: Video 1 Video 2 Video 3 Video 4 Video 5 Video 6

Question 6

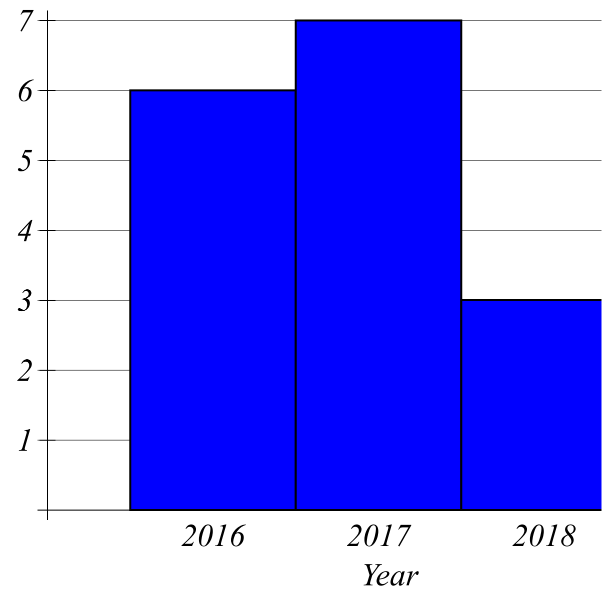
0/1 pt 3 19

The park service records the number of reported rattle snake sightings on Bald Mountain Trail and Camel Back Trail. The data gives rise to the following two bar graphs.

Rattlesnake Sightings on Bald Mountain Trail



Rattlesnake Sightings on Camel Bac



Use the bar graphs to fill in the contingency table below.

Reported Rattle Snake Sightings

	Year			
Trail	2016	2017	2018	Total
Bald Mountain				
Camel Back				
Total				

Question Help: Video

● **Question 7** 0/1 pt 3 19

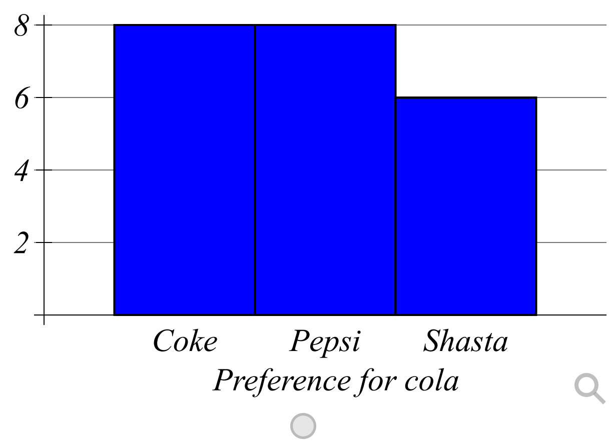
After surveying some of the classmates about their cola preferences, Vanessa came up with the following data set:

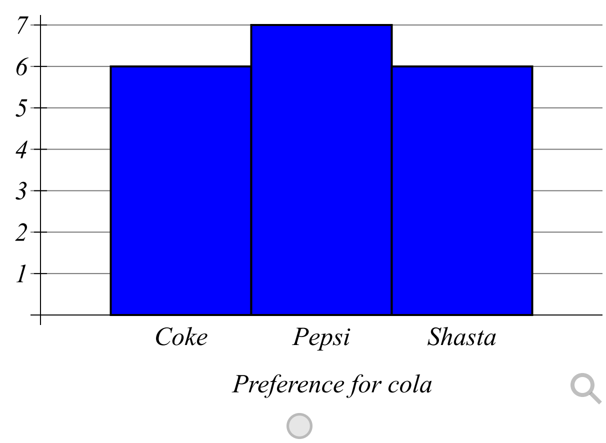
Coke	Pepsi	Coke	Shasta	Shasta
Shasta	Shasta	Shasta	Shasta	Coke
Coke	Pepsi	Pepsi	Pepsi	Pepsi
Coke	Shasta	Shasta	Shasta	Coke

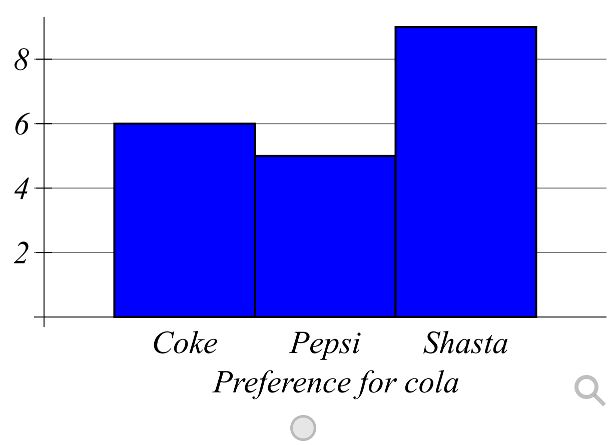
Construct a frequency and relative frequency distribution table (round the relative frequencies to 2 decimal places):

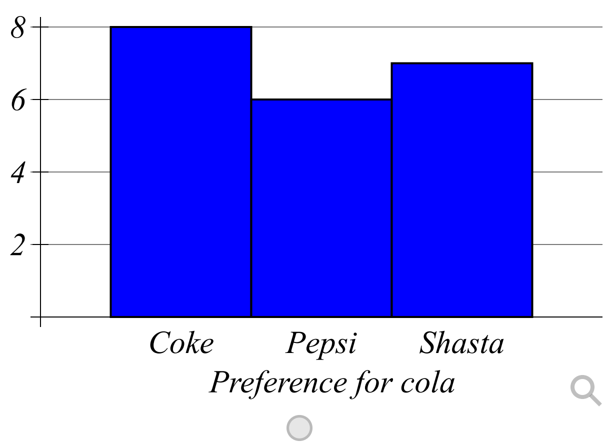
Type of cola	Frequency	Relative Frequency
Coke	______	______
Pepsi	______	______
Shasta	______	______
Total:	______	______

Construct a bar chart:









Construct a pie chart:

Question Help: Video

Question 8

0/1 pt 3 19

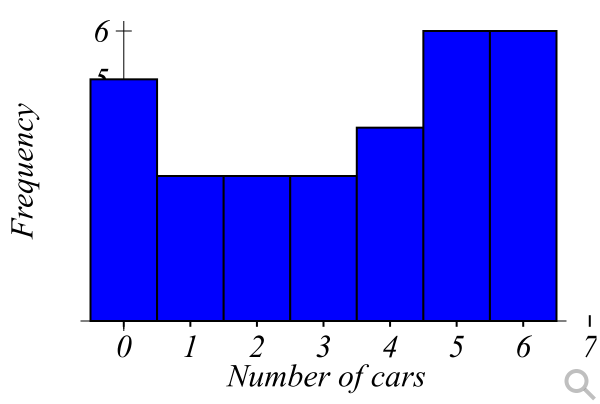
Nikole has surveyed their 30 classmates about the number of cars in their household and received the following data:

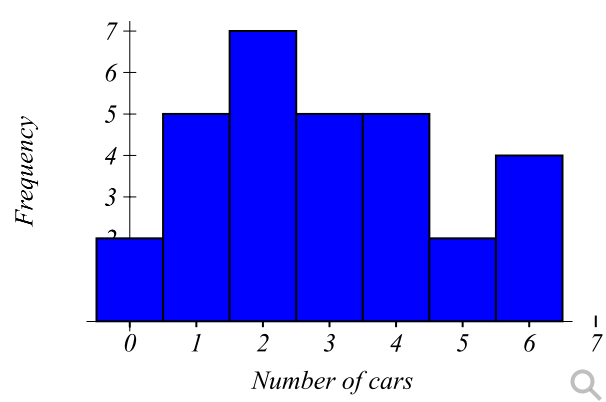
4	4	6	3	6
6	0	4	1	1
5	6	2	5	0
4	5	5	5	3
6	0	2	1	6
3	0	2	0	5

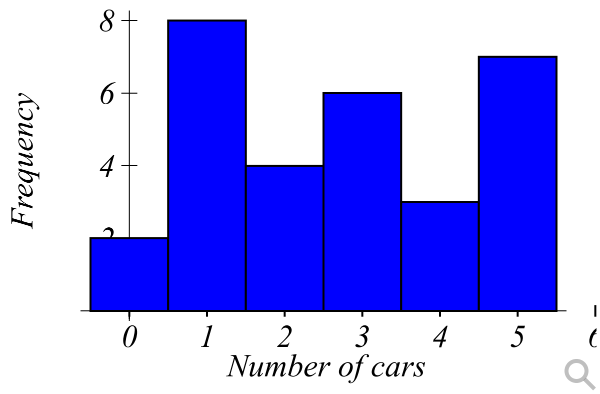
Construct a frequency and relative frequency distribution table: (Round relative frequences to 2 decimal places)

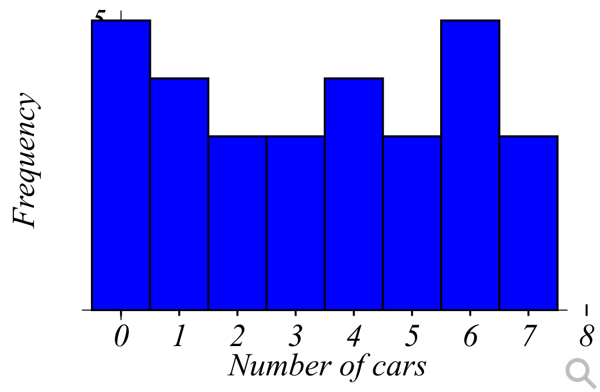
Number of cars	Frequency	Relative Frequency
0	________	________
1	________	________
2	________	________
3	________	________
4	________	________
5	________	________
6	________	________
Total:	________	________

Construct a histogram:









Question Help: Video

Question 9

0/1 pt 3 19

Mamadou obtained the data from the 35 most recent transaction at the store:

64.62	47.76	52.92	47.93	43.43
37.06	46.76	47.13	32.52	38.57
48.48	52.65	27.64	43.04	61.78
48.39	39.46	55.92	47.1	43.67
52.68	43.26	41.51	59.89	43.93
37.36	49.22	47.98	51.17	45.79
45.98	37.25	44.53	52.54	39.56

Use a grouping with the first class' lower limit 15 and the class width of 20 to:

- Create the frequency table:

Transaction values	Freq
`15 <= x < 35`	7
`35 <= x < 55`	20
`55 <= x < 75`	8

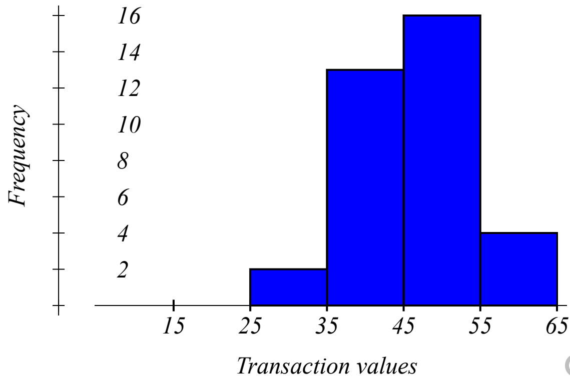
Transaction values	Freq
`20 <= x < 30`	1
`30 <= x < 40`	7
`40 <= x < 50`	18
`50 <= x < 60`	7
`60 <= x < 70`	2

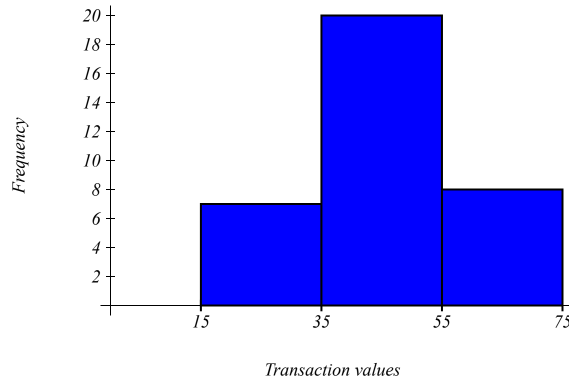
Transaction values	Freq
`15 <= x < 25`	0
`25 <= x < 35`	2
`35 <= x < 45`	13
`45 <= x < 55`	16
`55 <= x < 65`	4

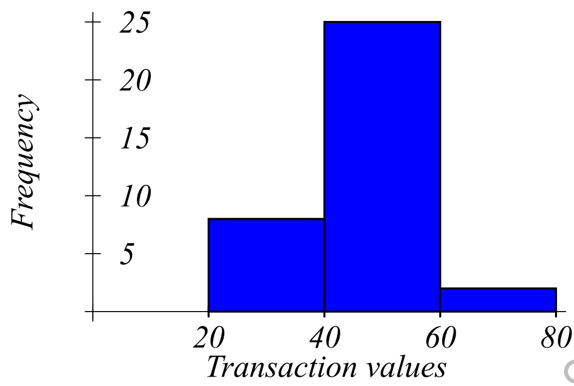
Transaction values	Freq
`15 <= x < 35`	2
`35 <= x < 55`	29
`55 <= x < 75`	4

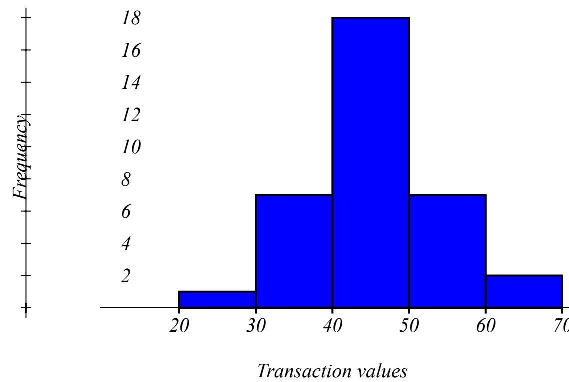
Transaction values	Freq
`20 <= x < 40`	8
`40 <= x < 60`	25
`60 <= x < 80`	2

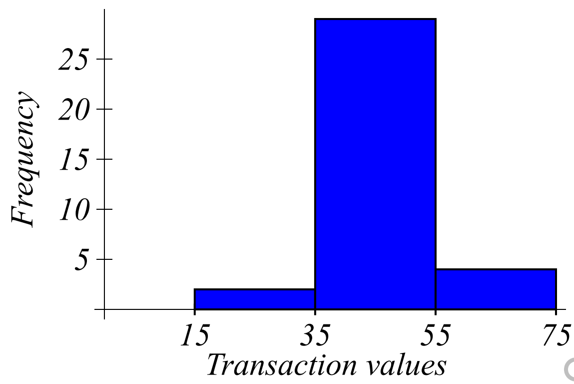
- Create the frequency histogram:











Identify the shape of the distribution of the data: Select an answer

Question Help: Video

Question 10 0/1 pt 3 19

Connor sampled the ages from the list of the Arizona's millionaires and observed the following data:

70	76	67	68	63
54	72	42	80	55
58	69	107	56	60
48	55	92	77	68
56	71	66	68	61
30	71	55	59	60
48	39	62	89	51
48	62	63	58	50

Use a grouping with the first class' lower limit 25 and the class width of 10 to:

- Create the frequency table:

Transaction values	Freq
`25 <= x < 35`	3
`35 <= x < 45`	3
`45 <= x < 55`	4
`55 <= x < 65`	3
`65 <= x < 75`	7
`75 <= x < 85`	7
`85 <= x < 95`	8
`95 <= x < 105`	3
`105 <= x < 115`	2

Transaction values	Freq
`30 <= x < 50`	6
`50 <= x < 70`	24
`70 <= x < 90`	8
`90 <= x < 110`	2

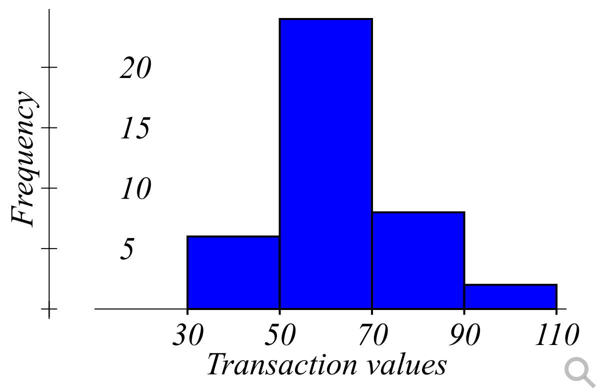
Transaction values	Freq
`30 <= x < 40`	2
`40 <= x < 50`	4
`50 <= x < 60`	11
`60 <= x < 70`	13
`70 <= x < 80`	6
`80 <= x < 90`	2
`90 <= x < 100`	1
`100 <= x < 110`	1

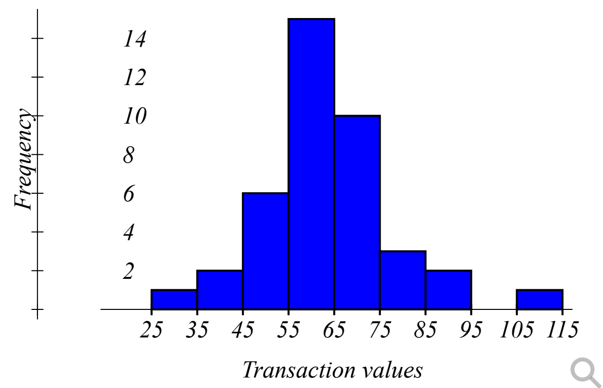
Transaction values	Freq
`25 <= x < 45`	3
`45 <= x < 65`	21
`65 <= x < 85`	13
`85 <= x < 105`	2
`105 <= x < 125`	1

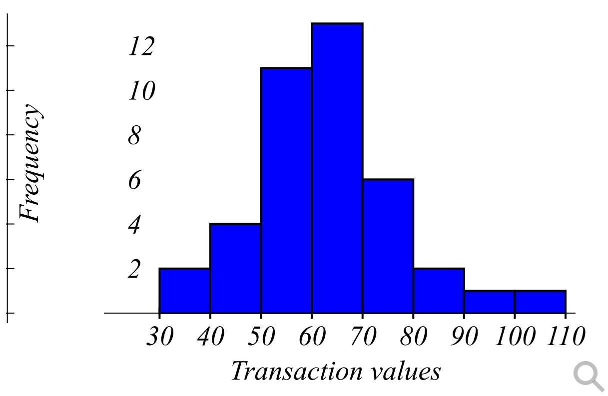
Transaction values	Freq
`25 <= x < 35`	1
`35 <= x < 45`	2
`45 <= x < 55`	6
`55 <= x < 65`	15
`65 <= x < 75`	10
`75 <= x < 85`	3
`85 <= x < 95`	2

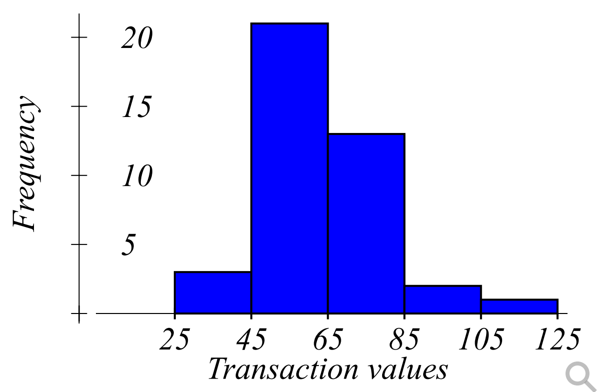
Transaction values	Freq
`95 <= x < 105`	0
`105 <= x < 115`	1

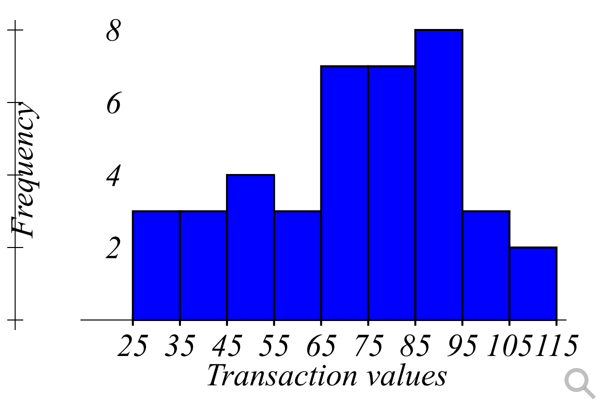
- Create the frequency histogram:



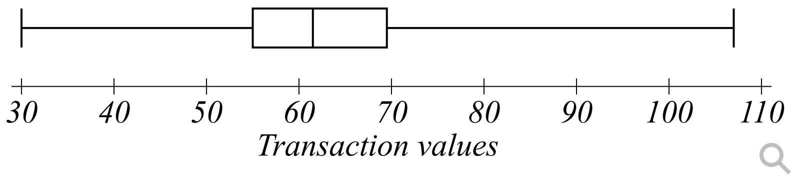


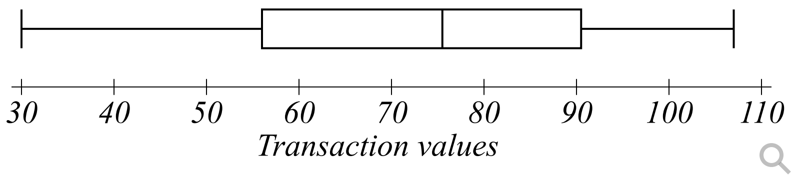






- Create the boxplot:





Identify the shape of the distribution of the data: Select an answer

Question Help: Video

Question 11 0/1 pt 3 19

If you are not familiar with Excel, watch the 1st "Help" video before doing this assignment. Skip any advertisements in the videos.

Topic: Emergency Room visits

1. Create an Excel file with the data shown in the 1st 3 columns of this attached file: 04_Excel_Basics_CSmith2_2.pdf [+]

2. In the 4th column use Excel's capability to add the numbers in the 2nd and 3rd columns.

For row 2, in cell D2 enter "=B2+C2". The sum should show up when you press "Enter".

For row 3, in cell D3 enter "=B3+C3".

Continue this pattern for the rest of the rows.

The 2nd help video shows how to enter an addition formula and how to copy it to other cells.

Your instructor may ask you to insert one or more charts. Follow your instructor's directions.

3. Save your Excel file with the name "Excel_Basics_MyName" (Replace "MyName" with your own name, so I can give you credit.)

4. Upload your Excel file in the editor box below. To attach a file:

1. Click the paperclip icon:
2. For the description, enter your file's name
3. In the Link box on the URL line, click the Browse icon at the right end of the line (a magnifying glass on Folder).
    a. Click the "Choose File" button and select the file from your computer you want to share
    b. Click the Upload Button
    c. The filename will appear at the top of the File Manager under "Just Uploaded". Click the file name to select it.
    d. Click "Insert" to insert your Excel file

6. Click "Submit"

Choose File No file chosen

Your Score will show up after your instructor grades your file. *Do not delete the file from the File Manager or else your instructor will not be able to access it.*

Question Help: Video 1 Video 2

Question 12

0/1 pt 3 19

Brandon surveyed 35 students and recorded the exact amounts they spent on books last semester:

167.33	170.39	124.4	171.87	165.07
170.9	98.52	109.82	79.71	189.43
187.74	93.46	158.78	158.31	74.32
172.74	91.6	170.07	210.68	146.23
192.1	178.89	122.49	166.35	175.59
141.87	127.04	138.98	98.47	177.96
157.54	182.76	175.77	149.42	118.26

Use a grouping with the first class' lower limit 70 and the class width of 20 to:

- Create the frequency table:

Transaction values	Freq
`65 <= x < 85`	2
`85 <= x < 105`	4
`105 <= x < 125`	4
`125 <= x < 145`	3
`145 <= x < 165`	5
`165 <= x < 185`	13
`185 <= x < 205`	3
`205 <= x < 225`	1

Transaction values	Freq
`70 <= x < 95`	4
`95 <= x < 120`	4
`120 <= x < 145`	5
`145 <= x < 170`	8
`170 <= x < 195`	13
`195 <= x < 220`	1

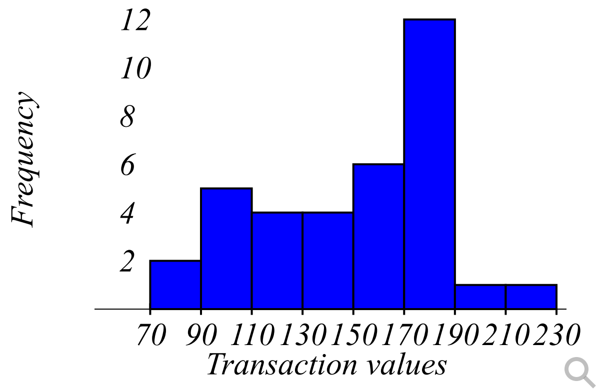
Transaction values	Freq
`70 <= x < 90`	2
`90 <= x < 110`	5
`110 <= x < 130`	4
`130 <= x < 150`	4
`150 <= x < 170`	6
`170 <= x < 190`	12
`190 <= x < 210`	1
`210 <= x < 230`	1

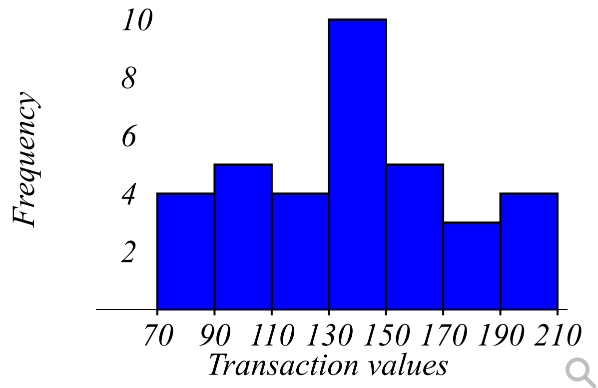
Transaction values	Freq
`70 <= x < 90`	4
`90 <= x < 110`	5
`110 <= x < 130`	4
`130 <= x < 150`	10
`150 <= x < 170`	5
`170 <= x < 190`	3

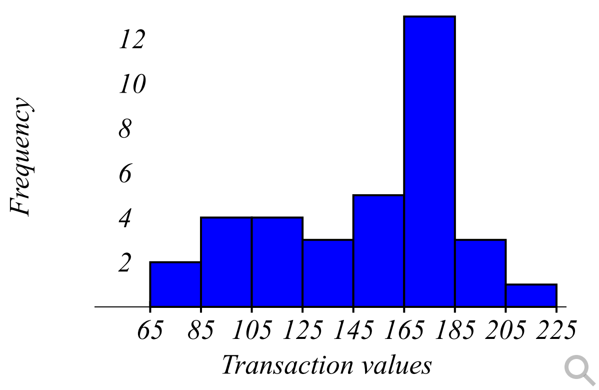
Transaction values	Freq
`190 <= x < 210`	4

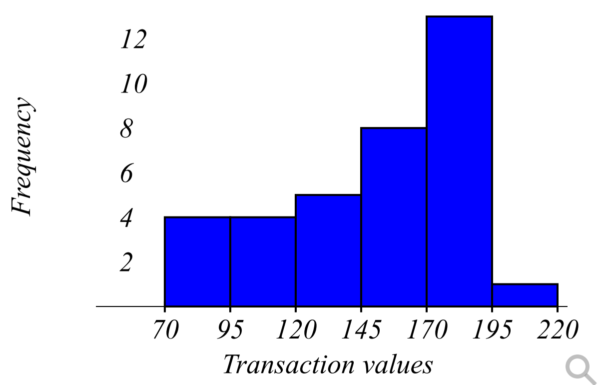
Transaction values	Freq
`65 <= x < 90`	2
`90 <= x < 115`	5
`115 <= x < 140`	5
`140 <= x < 165`	6
`165 <= x < 190`	15
`190 <= x < 215`	2

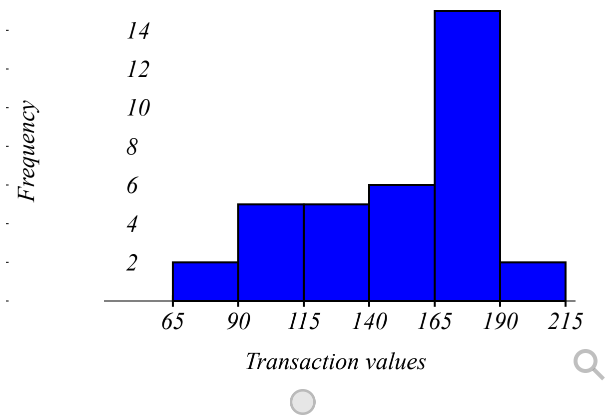
- Create the frequency histogram:











Identify the shape of the distribution of the data: Select an answer

Question Help: Video

Question 13

0/1 pt 3 19

Makenzie has surveyed their 25 classmates about the number of children under 18 in their household and received the following data:

2	4	0	5	5
3	2	0	6	1
4	4	4	1	2
6	2	3	0	6
4	3	0	0	2

Construct a frequency distribution table:

Number of children under 18	Frequency
0	______
1	______
2	______
3	______
4	______
5	______
6	______
Total:	______

Question Help: Video

Question 14

0/1 pt 3 19

Hunter has surveyed their 20 classmates about the number of cars in their household and received the following data:

0	4	6	6	3
6	2	6	3	6
2	4	6	6	1
1	0	5	2	6

Construct a cumulative frequency distribution table:

Number of cars	Frequency	Cumulative Frequency
0	2	______
1	2	______
2	3	______
3	2	______
4	2	______
5	1	______
6	8	______
Total:	20	

Question Help: Video

Question 15

0/1 pt 3 19

Corey surveyed the students in his neighborhood and obtained the following data:

Student	School level	School type
Student 1	Middle School	Public
Student 2	Middle School	Public
Student 3	High School	Public
Student 4	Elementary School	Public
Student 5	High School	Public
Student 6	High School	Private
Student 7	Middle School	Private
Student 8	Elementary School	Public
Student 9	High School	Private
Student 10	Elementary School	Public
Student 11	Middle School	Private
Student 12	Elementary School	Private
Student 13	High School	Private
Student 14	High School	Public
Student 15	Elementary School	Private
Student 16	Middle School	Private
Student 17	Middle School	Private
Student 18	Middle School	Public

Construct the contingency table that summarizes the school enrollment by level and type:

	Public	Private
Elementary School	______	______
Middle School	______	______
High School	______	______

Question Help: Video

Question 16 0/1 pt 3 19

Laura surveyed the students in her neighborhood and obtained the following data:

Student	School level	School type
Student 1	Middle School	Private
Student 2	High School	Public
Student 3	High School	Private
Student 4	Middle School	Public
Student 5	Elementary School	Public
Student 6	Elementary School	Private
Student 7	Middle School	Public
Student 8	High School	Public
Student 9	Middle School	Private
Student 10	High School	Private
Student 11	Middle School	Public
Student 12	Elementary School	Public
Student 13	High School	Private
Student 14	Middle School	Private
Student 15	Elementary School	Private
Student 16	High School	Private
Student 17	High School	Public
Student 18	Elementary School	Private
Student 19	Elementary School	Public
Student 20	High School	Public
Student 21	Middle School	Private
Student 22	Middle School	Public
Student 23	Elementary School	Private

Construct the contingency table that summarizes the school enrollment by level and type:

	Public	Private	Total
Elementary School	______	______	______
Middle School	______	______	______
High School	______	______	______
Total	______	______	______

Compute the totals for each row and column and the overall total.

Question Help: Video

Question 17 0/1 pt 3 19

The following are goals scored by a soccer team at each game in their recent season.

Complete the table by filling in the relative frequencies.

Goals	Frequency	Relative Frequency
0	6	
1	9	
2	4	
3	3	
4	2	
5	1	

Question Help: Video

Question 18

0/1 pt 3 19

A random sample of the body temperature of 25 young adults has the following distribution.

Complete the table by filling in the cumulative frequency distribution.

Temperature Group	Frequency	Cumulative Frequency
\`96.5-96.7\`	1	
\`96.8-97\`	3	
\`97.1-97.3\`	3	
\`97.4-97.6\`	5	
\`97.7-97.9\`	3	
\`98-98.2\`	6	
\`98.3-98.5\`	2	
\`98.6-98.8\`	2	

Question Help: Video

Question 19

0/1 pt 3 19

Fifty randomly selelected individuals reported their average monthly energy bills and their results are given in the following frequency distribution. Complete the table by filling in the midpoint of each class range.

Utility Bills	Midpoint	Frequency
40 - 50		12
51 - 61		7
62 - 72		12
73 - 83		9
84 - 94		10

Question Help: Video

Question 20

0/1 pt 3 19

A randomly selelected group of individuals were asked how many marbles they believed were contained in a sealed jar. The following frequency distribution summarizes the results. Complete the table by determining the midpoints, relative frequencies, and cumulative frequencies.

Marbles	Frequency	Midpoints	Relative Frequ
180 - 184	7		
185 - 189	4		
190 - 194	2		
195 - 199	11		
200 - 204	3		

Question Help: Video

Question 21

0/1 pt 3 19

All 8th grade students at a school answered Yes or No to the two survey questions shown.

- Have you ridden on a city bus in the last year? Yes No
- Have you taken an uber in the last year? Yes No

The same students responded to both questions. Complete the two-way frequency table to show the correct totals for the given data.

	Yes Uber	No Uber	Total
Yes Bus	35	____	115
No Bus	65	155	____
Total	____	235	____

Question Help: Video

Question 22 0/1 pt 3 19

The ages for 30 lottery winners was recorded and a frequency distribution of this data is given below:

Age	Frequency
`20 - 32`	3
`33 - 45`	5
`46 - 58`	15
`59 - 71`	2
`72 - 84`	1
`85 - 97`	4

What is the class width of this distribution?

____

What is the frequency of lottery winners of age between 46 and 58?

____

What is the frequency of lottery winners between the ages of 20 and 45?

____

What is the midpoint of the third class?

____

Rounded to three decimal places, what is the relative frequency of the fourth class?

____

What is the cumulative frequency of the third class?

____

What is the upper class limit of the second class?

____

What is the lower class boundary of the fifth class?

____

Question Help: Video

Question 23

0/1 pt 3 19

Twenty movies were released over a particular summer and their MPAA ratings (G, PG, PG-13, R) are given in the table below.

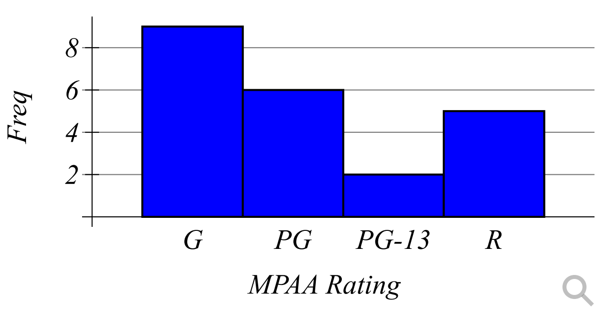
G	G	G	PG	PG
PG	R	R	G	G
PG	PG	G	R	R
G	R	G	PG	G

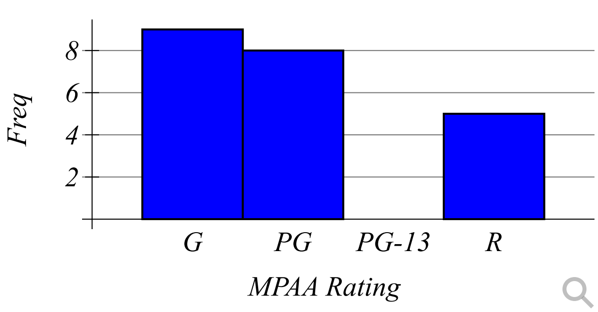
---

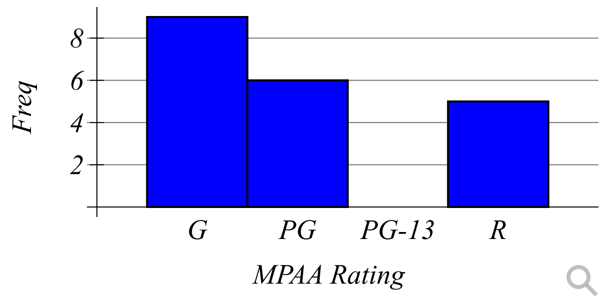
a) Complete the frequency and relative frequency distribution for this qualitative data set.

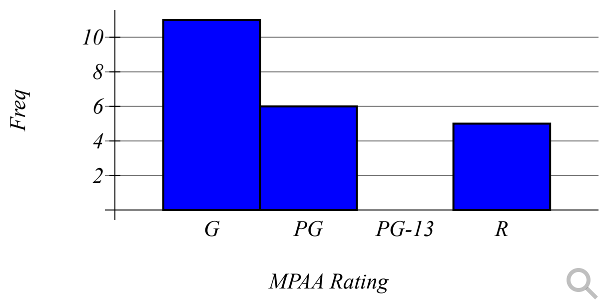
MPAA Rating	Frequency	Relative Frequency
G		
PG		
PG-13		
R		

b) Which of the following is the correct histogram for the distribution?









Question Help: Video

Question 24

0/1 pt 3 19

Calculate the midpoints and relative frequencies for each class of the frequency table.

Round your answers to three decimal places, if necessary.

Test Scores	Frequency	Midpoints	Relative Frequency
40 - 49	8		
50 - 59	4		
60 - 69	10		
70 - 79	5		
80 - 89	9		
90 - 99	1		

Question Help: Video

Question 25

0/1 pt 3 19

To analyze how Arizona workers ages 16 or older travel to work (carpool, private vehicle, public transportation, other), data from 20 workers were collected and are shown in the table below.

Carpool
Carpool
Carpool
Private
Private
Private
Public
Other
Carpool
Private
Other
Private
Private
Public
Other
Private
Carpool
Public
Public
Other

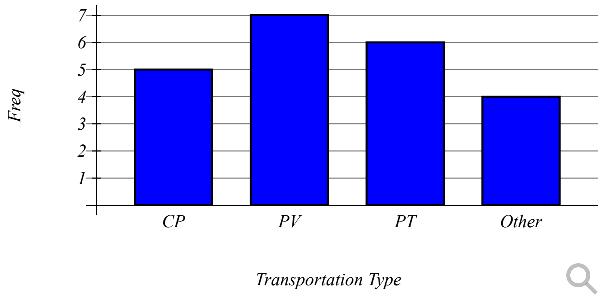
---

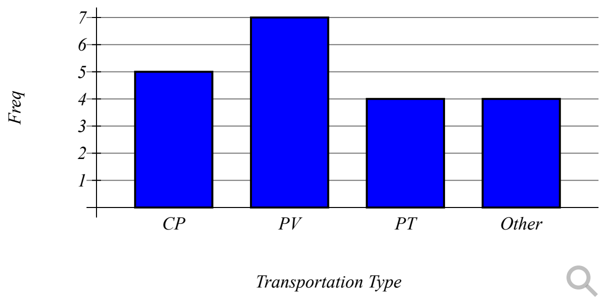
a) Since data were collected for [?] [Select an answer] variable(s), the correct graph to make is a [Select an answer] .

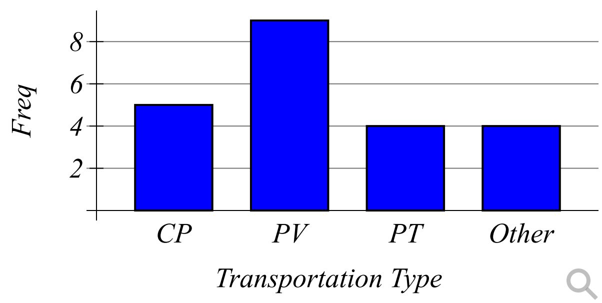
b) Complete the frequency/relative frequency table.

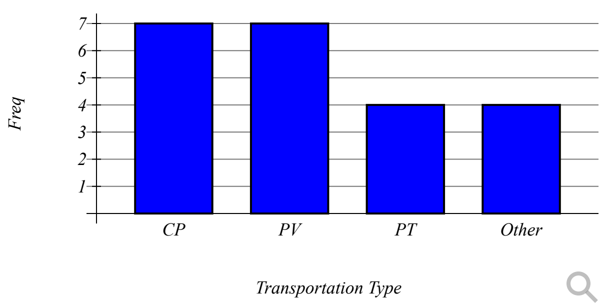
Transportation Type	Frequency	Relative Frequency
Carpool		
Private		
Public		
Other		

c) Which of the following is the correct bar chart for the given data?









d) Which of the following is the correct pie chart for the given data?

Question Help: Video 1 Video 2 Video 3 Video 4 Video 5 Video 6

Question 26

0/1 pt 3 19

To analyze how Arizona workers ages 16 or older travel to work (carpool, private vehicle, public transportation, other), data from 15 workers were collected and are shown in the table below.

Carpool
Carpool
Carpool
Private
Private
Private
Public
Other
Other
Private
Private
Private
Public
Private
Other

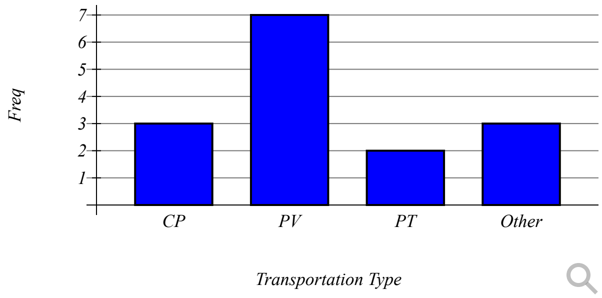
---

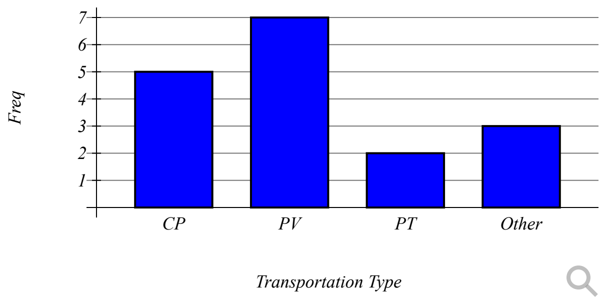
a) Since data were collected for ? Select an answer variable(s), the correct graph to make is a Select an answer .

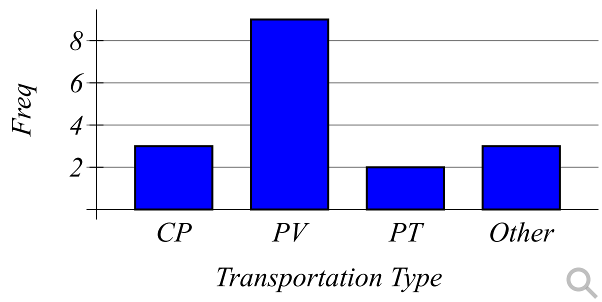
b) Complete the frequency/relative frequency table. (Hint: For the relative frequency, use reduced fractions)

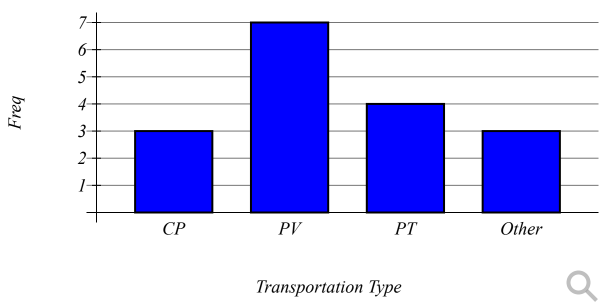
Transportation Type	**Frequency**	**Relative Frequency**
Carpool		
Private		
Public		
Other		

c) Which of the following is the correct bar chart for the given data?









d) Which of the following is the correct pie chart for the given data?

Question Help: Video 1 Video 2 Video 3 Video 4 Video 5 Video 6

Question 27 0/1 pt 3 19

The density of people per square kilometer for African countries is in the table below. ('Density of people,' 2013).

26
42
16
105
157
39
123
18
341
134
79
45
3
51
415
8
81
12
4
65
176
563
12
2
367
31
194
29
9
3

---

a) Since data were collected for ? Select an answer variable(s), the correct graph to make is a Select an answer .

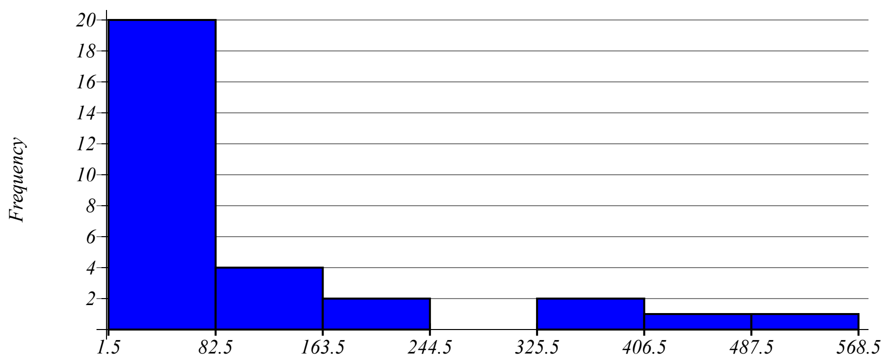
b) Using the formula class width = (maxX - minX)/(number of classes) and rounding up to the next whole number, the class width is ________

Complete the frequency/relative frequency table using 7 classes.

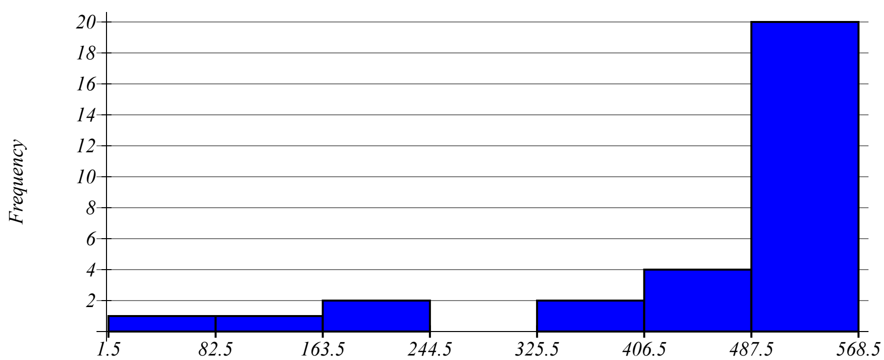
Round relative frequencies to 3 decimal places.

Classes	Frequency	Relative Frequency
1.5 - ____	____	____
____ - ____	____	____
____ - ____	____	____
____ - ____	____	____
____ - ____	____	____
____ - ____	____	____
____ - ____	____	____

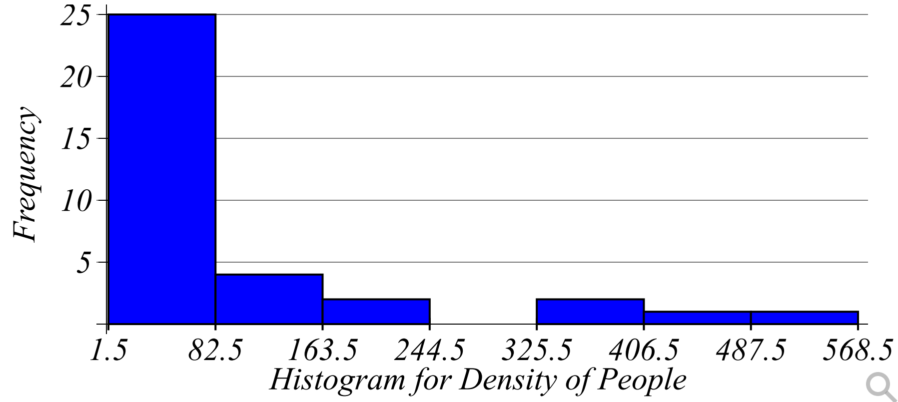
c) Which of the following is the correct histogram for the given data?

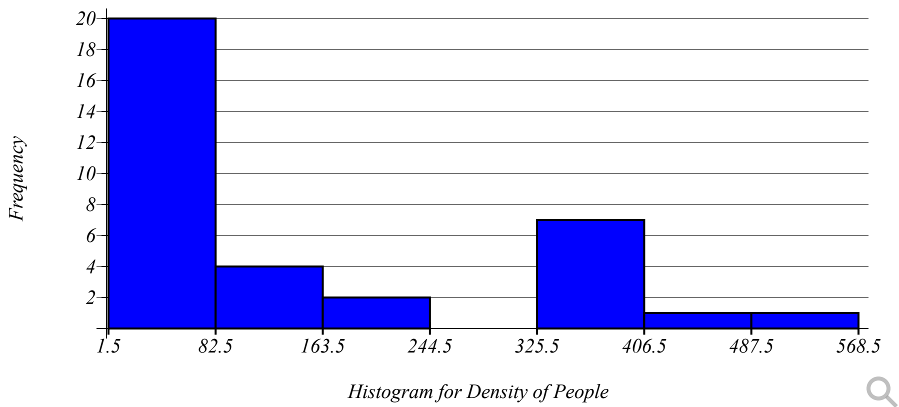


Histogram for Density of People



Histogram for Density of People





Question Help: Video

Question 28

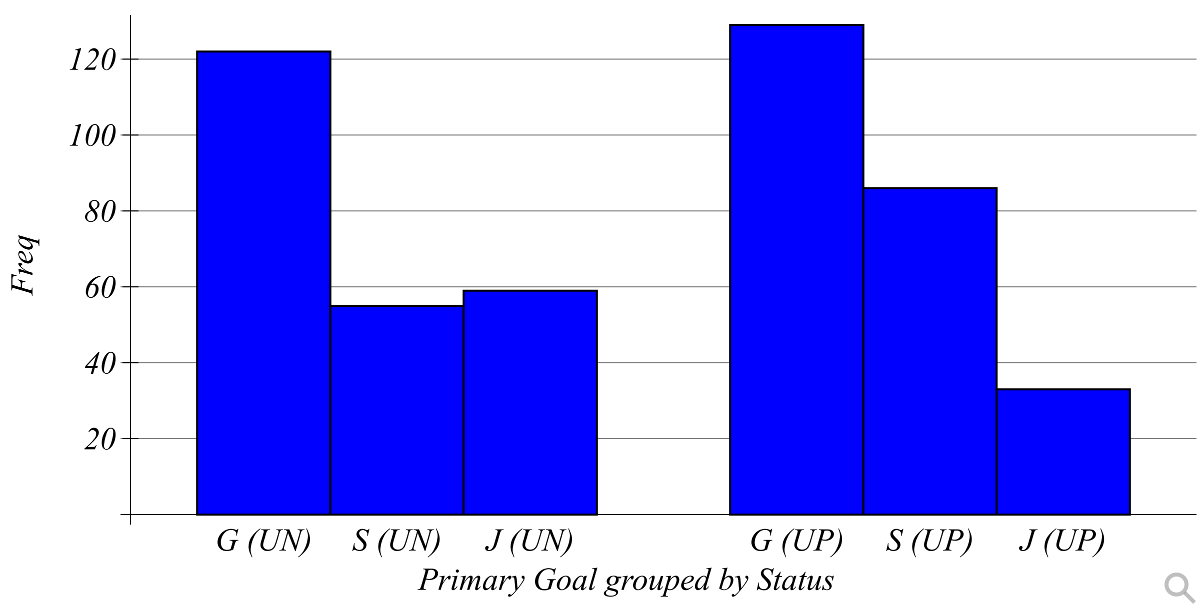
0/1 pt 3 19

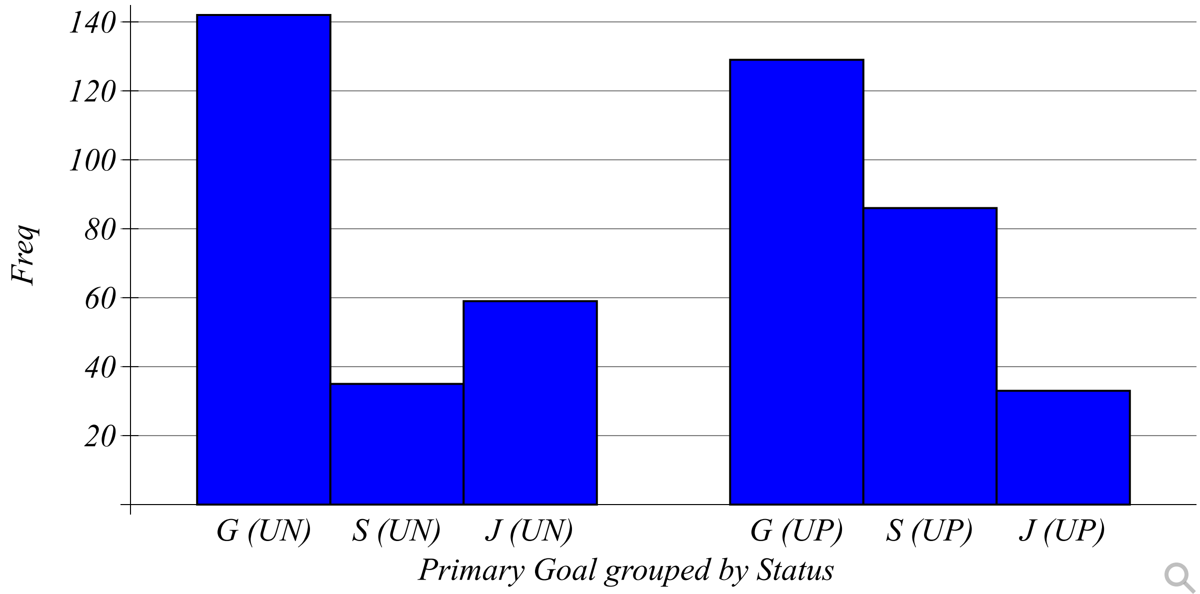
In a study of high school age students, the following data were collected on their status and their primary goal.

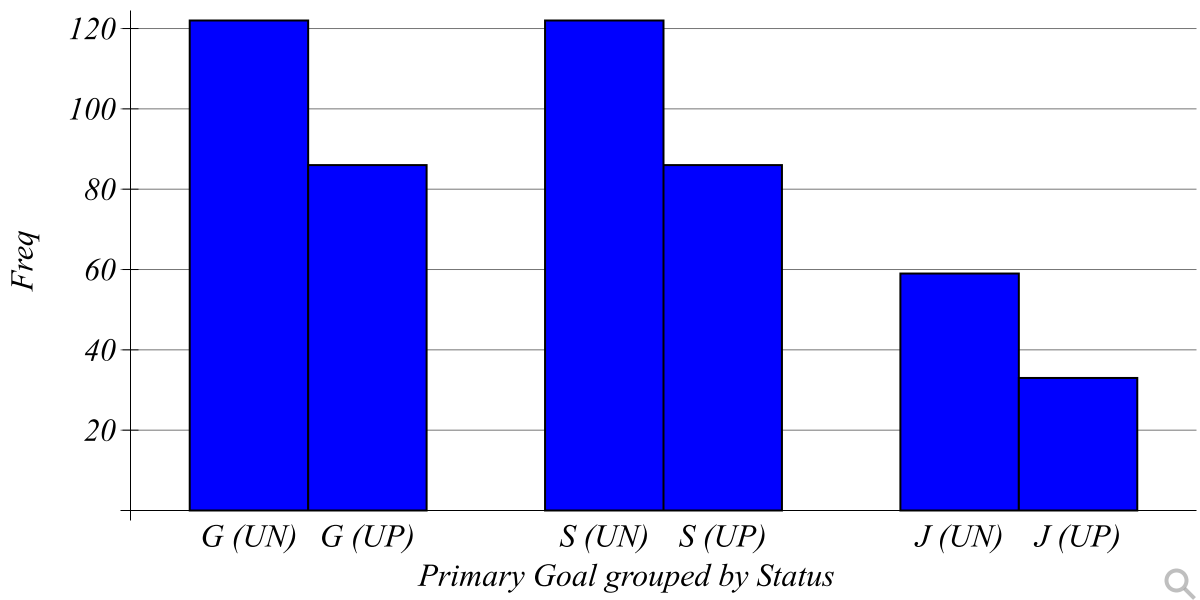
		Primary Goal		
Status	College Preparation (C)	Sports and Activities (S)	Job/Employment (J)	Row Totals
UnderClassman (UN)	122	55	59	236
UpperClassman (UP)	129	86	33	248
Column Totals	251	141	92	484

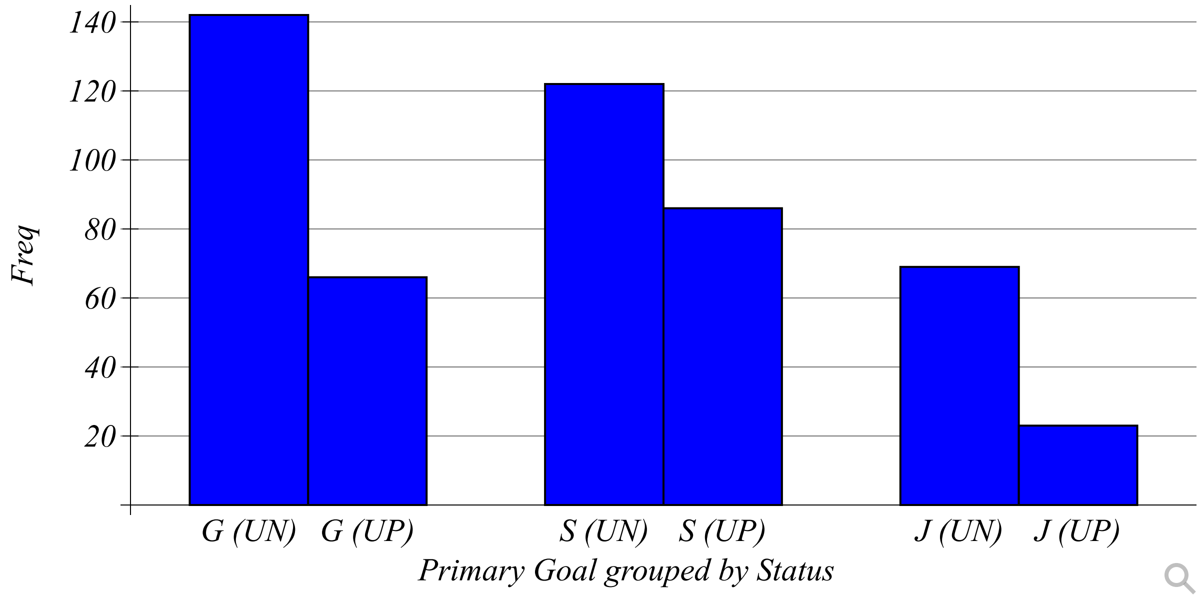
a) Since data were collected for [?] [Select an answer] variable(s), the correct graph to make is a [Select an answer]

b) Which of the following is the correct graph for the above data with Primary Goal grouped by Status.




140
120
100
80
60
40
20
Freq
G (UN)
S (UN)
J (UN)
G (UP)
S (UP)
J (UP)
Primary Goal grouped by Status


120
100
80
60
40
20
Freq
G (UN)
G (UP)
S (UN)
S (UP)
J (UN)
J (UP)
Primary Goal grouped by Status

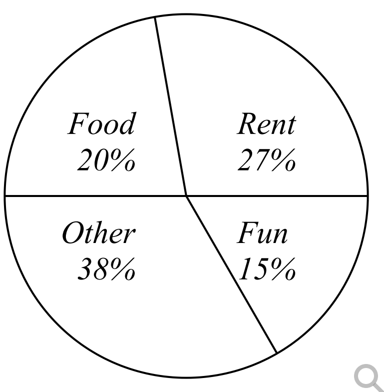


Question Help: Video 1 Video 2 Video 3 Video 4 Video 5 Video 6

Question 29

0/1 pt 3 19

Lillian categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The percents she spent in each category are pictured here.



If Lillian spent a total of $2900 this month, how much did she spend on Fun?

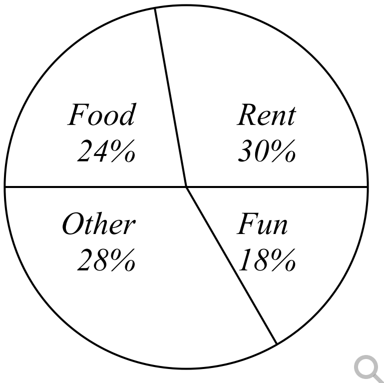
$ ____

Question Help: Video

Question 30

0/1 pt 3 19

Tabria categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The percents she spent in each category are pictured here.



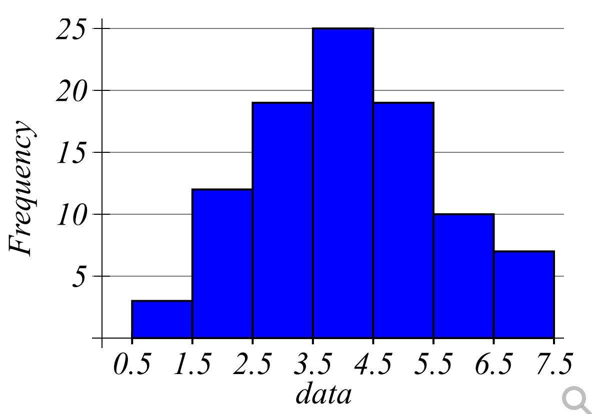
If Tabria spent a total of $2300 this month, how much did she spend on Fun?

- $18
- $414
- $437

Question Help: Video

Question 31 0/1 pt 3 19

Determine the distribution of the data pictured below



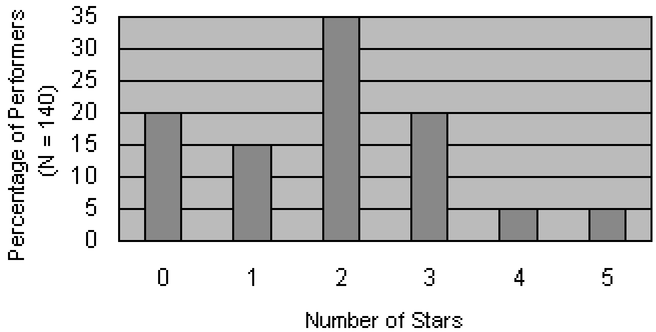
- Uniform
- Bell-shaped
- Skewed-right
- Skewed-left

Question Help: Video

Question 32

0/1 pt 3 19

The data in the figure below represents the number of stars earned by 140 performers in a talent competition.



How many performers earned exactly 2 stars? (be careful - compare the question wording to the scale on the graph)

Question Help: Video

Question 33

0/1 pt 3 19

In a relatively pointless survey, the following data was collected on cars in a Costco parking lot: there were 27 black cars; 100 purple cars; 25 pink cars; 42 white cars; 35 light blue cars; 19 cars of some other color.

If you were to construct a Pareto chart, how would the second bar in the graph be labeled? Select an answer

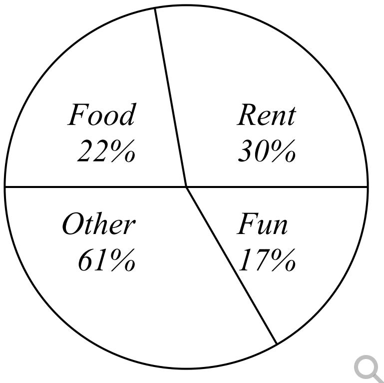
If you were to construct a Pareto chart, what is the frequency of the first bar?

Question Help: Video

Question 34

0/1 pt 3 19

Makenzie categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The percents she spent in each category are pictured here.



If Makenzie spent a total of $2700 this month, how much did she spend on Rent?

$ ________

Question Help: Video

Question 35

0/1 pt 3 19

Here is a data set summarized as a stem-and-leaf plot:

3	00022222579
4	01122344466688
5	04455
6	499

How many data values are in this data set? `n=` ________

What is the minimum value in the last class? ________

What is the minimum value in the entire sample? ________

How many of the original values are greater than 40? ________

Question Help: Video

Question 36

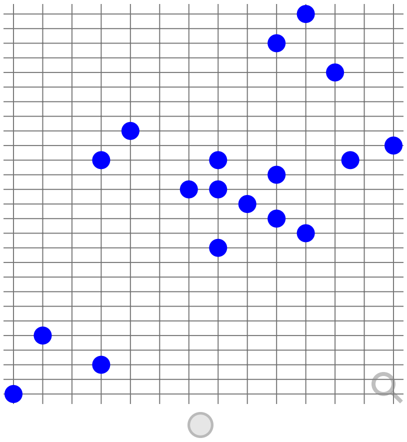
0/1 pt 3 19

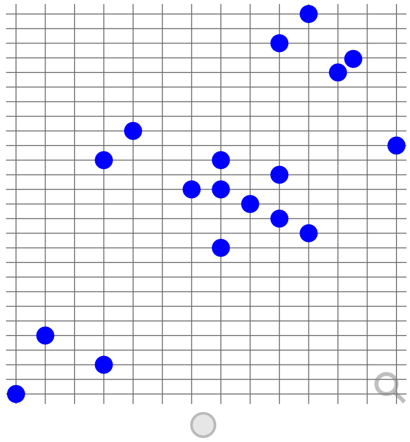
When an anthropologist finds skeletal remains, they need to figure out the height of the person. The height of a person (in cm) and the length of their metacarpal bone (in cm) were collected for 18 sets of skeletal remains. The data are in the table below.

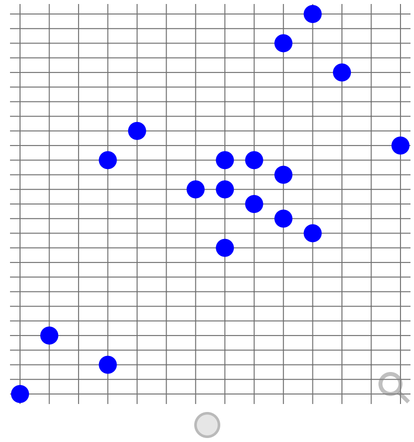
length of metacarpal	height
46	175
45	173
46	173
48	174
46	169
47	175
49	170
48	183
52	176
49	185
50	181
42	161
48	171
40	163
42	175
39	159
47	172
43	177

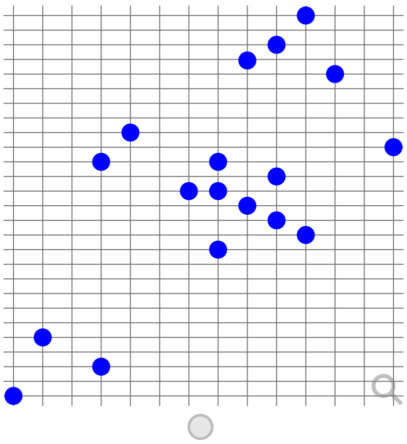
a) Since data were collected for [?] [Select an answer] variable(s), the correct graph to make is a [Select an answer] .

b) Make a scatterplot of X versus Y in StatCrunch or on your TI84. Which of the following is the correct graph?









c) Which of the following is true about the relation between length of metacarpal and height based on the graph above?

Select an answer

Question Help: Video 1 Video 2

Question 37

0/1 pt 3 19

The data represents the price (in cents per pound) paid to 21 farmers for apples.

22.4	23.5	22.8	25.1	24.1
23.6	23.4	24.1	27.2	24.6
22.1	22.4	22.7	22.1	22.6
23.5	25.9	22	23.4	23.2
24.4				

---

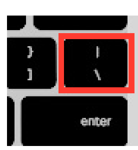
1) **Complete the stem-and-leaf plot below.**

*** You must put commas between the **SORTED leaves least to greatest**.*
*** Type **"DNE"** if the leaves don't exist.*

Stems	Leaves

Key: We'll make a key to explain how the stem and leaf plot works, so type 18|0 in this box.

Key: ________ = 18.0

Answer above → (Shift  *) for the separator bar with **NO Spaces**.*

2) **What is the SOCS of the data?**

a. **Shape:** Select an answer

b. **Outlier(s):** ?

c. **Center:** ______

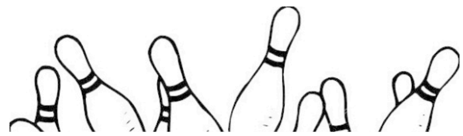
d. **Spread:** ______ ←*Use a* ***dash*** *in your answer with* ***No Spaces*****.**

Question Help: Video

Question 38

0/1 pt 3 19

Katie recorded her bowling scores over the last 19 games. By looking at her data, she was able to get a better picture of her performances. Make a stem-leaf plot of her bowling scores and answer the questions:



152	135	152	145	130
154	151	120	142	120
150	141	143	149	140
131	141	130	101	

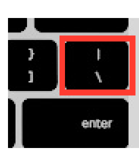
---

1) **Complete the stem-and-leaf plot below.**

*** You must put commas between the **SORTED leaves least to greatest**.*
*** Type **"DNE"** if the leaves don't exist.*

Stems	Leaves

Key: ________ = 123

Answer above → (Shift 

*) for the separator bar with **NO Spaces**.*

2) **What is the SOCS of the data?**

a. **Shape:** Select an answer

b. **Outlier(s):** ?

c. **Center:** ______

d. **Spread:** ______ ←*Use a* ***dash*** *in your answer with* ***No Spaces.***

Question Help: Video

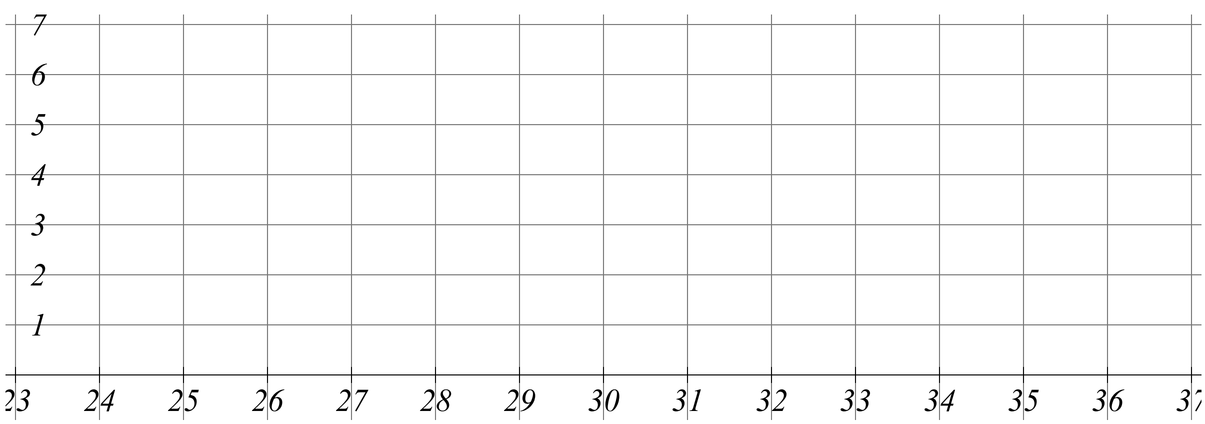
Question 39

0/1 pt 3 19

The following data set outlines the scores students had on an Algebra test.

29	32	27	30	31
25	26	34	34	33
27	28	26	33	25
33	28	34	34	34

Draw a dot plot of this data on the graph below.



Question Help: Video

Question 40

0/1 pt 3 19

The stem and leaf plot below represents the ages in years of family members attending a large family picnic.

Ages:

0 | 3 9
1 | 0 2 4 8
2 | 3 5 6
3 | 2 2 2 5
4 | 7 7

Key: 2 | 3 means 23 years

Assuming the data represents a population, answer the following.
Round answers to two decimal places, if necessary:

Mean = ______

Median = ______

Mode = ______

Range = ______

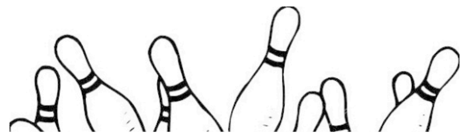
Which shape fits the distribution best? Select an answer

Question Help: Video

Question 41

0/1 pt 3 19

Katie recorded her bowling scores over the last 19 games. By looking at her data, she was able to get a better picture of her performances. Make a stem-leaf plot of her bowling scores and answer the questions:



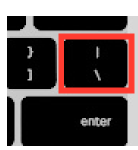
141	147	156	129	137	104
156	152	141	125	141	141

Complete the stem-and-leaf plot below.

*** You must put commas between the* ***SORTED leaves least to greatest****.*
*** Type* ***"DNE"*** *if the leaves don't exist.*

Stems	Leaves

Key: ______ = 123

Answer above → (Shift  *) for the separator bar with* ***NO Spaces****.*

Question Help: Video 1 Video 2

Question 42 0/1 pt 3 19

The table below gives the number of hours spent watching TV last week by a sample of 24 children. Find the minimum, maximum, range, mean, and standard deviation of the following data using using Excel, a calculator or other technology. The minimum, maximum, range should be entered as exact values. Round the mean and standard deviation to two decimals places.

76	82	59	62	44	42
56	52	42	86	34	87
60	69	35	56	81	64
53	66	38	56	94	19

Min = ______ Max = ______ Range = ______

Mean = ______ Standard Deviation = ______

Question Help: Video

Question 43

0/1 pt 3 19

The physical plant at the main campus of a large state university recieves daily requests to replace florecent lightbulbs. The distribution of the number of daily requests is bell-shaped and has a mean of 52 and a standard deviation of 8. Using the Empirical Rule rule, what is the approximate percentage of lightbulb replacement requests numbering between 44 and 52?

Do not enter the percent symbol.
ans = ______ %

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Question 44

0/1 pt 3 19

A survey collected the data below.

x
30
2
1
24
17

The mean is 14.8. Calculate the deviation of `x=17`. That is, find `17-14.8`.

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Question 45

0/1 pt 3 19

We are going to calculate the standard deviation for the following set of data.

1) First, calculate the mean.

`\mu = ` ________

2) Fill in the table below. Fill in the differences of each data value from the mean, then the squared differences.
3) Calculate the standard deviation.

`x`	`x-\mu`	`(x-\mu)^2`
14		
14		
14		
12		
11		
	Total	

Standard deviation:

`\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}} = ` ________ *Round to two decimal places*

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